

SCTEP problem formulation

Input data:

- Network data
- Set of contingencies
- Set of scenarios
- Flexibility data
- RES data
- Investment options
- Costs and penalties

**AC SCOPF
model**
(1a)-(1l)

**Objective
function**
(2) or (3)

Analysis of the optimal SCTEP solution
Screening of the investment options

Coalitional analysis

Cooperative game formulation:
Defining the set of players and colations

```
for coalition S in set do
  solve SCTEP model (1a)-(1l), (2)
  calculate the value of coalition  $v(S)$ 
end for
for player I in N do
end for
```

Compute the **Shapley value** for each
player i (5)

Output

Analysis of the
distribution of
marginal contributions
Investments with the
highest synergistic
capabilities
Interpretation of the
value of flexibility